JEE Mains 2019 Chapter wise Question Bank

Complex Numbers – Questions

Q1

Let $A=\left\{\theta \in\left(-\frac{\pi}{2}, \pi\right): \frac{3+2 i \sin \theta}{1-2 i \sin \theta} \text{ is purely imaginary}\right\}$.

Then the sum of the elements in A is:

(1) $\frac{5\pi}{6}$ (2) π

(3) $\frac{3\pi}{4}$ (4) $\frac{2\pi}{3}$

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Q2

Let z_0 be a root of the quadratic equation, $x^2+x+1=0$. If $z=3+6i z_0^{81}-3i z_0^{93}$, then arg z is equal to:

(1) $\frac{\pi}{4}$ (2) $\frac{\pi}{6}$

(3) $\frac{\pi}{3}$ (4) 0

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Q3

Let z_1 and z_2 be any two non-zero complex numbers such that $3|z_1| = 4|z_2|$. If $z=\frac{3z_1}{2z_2}+\frac{2z_2}{3z_1}$ then:

(1) $\text{Re}(z)=0$ (2) $|z|=\sqrt{\frac{5}{2}}$

(3) $|z|=\frac{1}{2}\sqrt{\frac{17}{2}}$ (4) $\text{Im}(z)=0$

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Q4

Let $z=\left(\frac{\sqrt{3}}{2}+\frac{i}{2}\right)^5+\left(\frac{\sqrt{3}}{2}-\frac{i}{2}\right)^5$. If $R(z)$ and $I(z)$ respectively denote the real and imaginary parts of z, then:

(1) $I(z)=0$

(2) $R(z)>0$ and $I(z)>0$

(3) $R(z)<0$ and $I(z)>0$

(4) $R(z)=-(3)$

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Q5

Let $\left(-2-\frac{1}{3}i\right)^3=\frac{x+iy}{27}\left(i=\sqrt{-1}\right)$, where x and y are real numbers then $y-x$ equals :

(1) 91 (2) -85

(3) 85 (4) -91

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Q6

Let z be a complex number such that $|z|+z=3+i$ $\left(\text{where } i=\sqrt{-1}\right)$.

Then $|z|$ is equal to :

(1) $\frac{\sqrt{34}}{3}$ (2) $\frac{5}{3}$

(3) $\frac{\sqrt{41}}{4}$ (4) $\frac{5}{4}$

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Q7

If $\frac{z-\alpha}{z+\alpha}(\alpha \in R)$ is a purely imaginary number and $|z| = 2$, then a value of α is :

(1) 2 (2) 1

(3) $\frac{1}{2}$ (4) $\sqrt{2}$

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Q8

Let z_1 and z_2 be two complex numbers satisfying $|z_1| = 9$ and $|z_2| - |3|-|4i||=|4$. Then the minimum value of $|z_1 - z_2|$ is :

(1) 0 (2) $\sqrt{2}$

(3) 1 (4) 2

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Q9

If $z = \frac{\sqrt{3}}{2} + \frac{i}{2}(i = \sqrt{-1})$, then $(1 + iz + z^5 + iz^8)^9$ is equal to:

(1) 0 (2) 1

(3) $(-1 + 2i)^9$ (4) -1

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Q10

All the points in the set $S = \left\{\frac{\alpha+i}{\alpha-i} : \alpha \in R\right\}(i = \sqrt{-1})$ lie on a:

(1) straight line whose slope is 1.
(2) circle whose radius is 1.
(3) circle whose radius is $\sqrt{2}$.
(4) straight line whose slope is -1.

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Q11

Let $z \in C$ be such that $|z| < 1$. If $\omega = \frac{5+3z}{5(1-z)}$, then :

(1) $5\,\mathrm{Re}(\omega) > 4$ (2) $4\,\mathrm{Im}(\omega) > 5$

(3) $5\,\mathrm{Re}(\omega) > 1$ (4) $5\,\mathrm{Im}(\omega) < 1$

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Q12

If $a > 0$ and $z = \frac{(1+i)^2}{a-i}$, has magnitude $\sqrt{\frac{2}{5}}$, then $\bar{z}$ is equal to :

(1) $-\frac{1}{5} - \frac{3}{5}i$ (2) $-\frac{3}{5} - \frac{1}{5}i$

(3) $\frac{1}{5} - \frac{3}{5}i$ (4) $-\frac{1}{5} + \frac{3}{5}i$

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Q13

If z and ω are two complex numbers such that $|z\omega| = 1$ and $\arg(z) - \arg(\omega) = \frac{\pi}{2}$, then:

(1) $\bar{z}\omega = i$ (2) $z\bar{\omega} = \frac{-1+i}{\sqrt{2}}$

(3) $\bar{z}\omega = -i$ (4) $z\bar{\omega} = \frac{1-i}{\sqrt{2}}$

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Q14

The equation $|z - i| = |z - 1|, i = \sqrt{-1}$, represents:

(1) a circle of radius $\frac{1}{2}$.
(2) the line through the origin with slope 1.
(3) a circle of radius 1.
(4) the line through the origin with slope -1.

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Q15

Let $z \in C$ with $\mathrm{Im}(z) = 10$ and it satisfies $\frac{2z-n}{2z+n} = 2i - 1$ for some natural number n. Then :

(1) $n = 20$ and $\mathrm{Re}(z) = -10$
(2) $n = 40$ and $\mathrm{Re}(z) = 10$
(3) $n = 40$ and $\mathrm{Re}(z) = -10$
(4) $n = 20$ and $\mathrm{Re}(z) = 10$

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JEE Mains 2019 Chapter wise Question Bank

Complex Numbers - Solutions

Q1

(4) Suppose $z = \dfrac{3+2i\sin\theta}{1-2i\sin\theta}$

Since, z is purely imaginary, then $z + \bar{z} = 0$

$$\Rightarrow \frac{3+2i\sin\theta}{1-2i\sin\theta} + \frac{3-2i\sin\theta}{1+2i\sin\theta} = 0$$

$$\Rightarrow \frac{(3+2i\sin\theta)(1+2i\sin\theta)+(3-2i\sin\theta)(1-2i\sin\theta)}{1+4\sin^2\theta} = 0$$

$$\Rightarrow \sin^2\theta = \frac{3}{4} \quad \Rightarrow \quad \sin\theta = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = -\frac{\pi}{3}, \frac{\pi}{3}, \frac{2\pi}{3}$$

Now, the sum of elements in $A = -\dfrac{\pi}{3} + \dfrac{\pi}{3} + \dfrac{2\pi}{3} = \dfrac{2\pi}{3}$

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Q2

(1) $\because$ z_0 is a root of quadratic equation

$x^2 + x + 1 = 0$

$\therefore \quad z_0 = \omega$ or $\omega^2 \Rightarrow z_0^3 = 1$

$\therefore \quad z = 3 + 6i\,z_0^{81} - 3i\,z_0^{93}$

$= 3 + 6i((z_0)^3)^{27} - 3i((z_0)^3)^{31}$

$= 3 + 6i - 3i$

$= 3 + 3i$

$\therefore \quad \arg(z)\tan^{-1}\left(\dfrac{3}{3}\right) = \dfrac{\pi}{4}$

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Q3

(Bonus)

Let $z_1 = r_1 e^{i\theta}$ and $z_2 = r_2 e^{i\phi}$

$3|z_1| = 4|z_2| \Rightarrow 3r_1 = 4r_2$

$$z = \frac{3z_1}{2z_2} + \frac{2z_2}{3z_1} = \frac{3r_1}{2r_2}e^{i(\theta-\phi)} + \frac{2}{3}\frac{r_2}{r_1}e^{i(\phi-\theta)}$$

$$= \frac{3}{2} \times \frac{4}{3}(\cos(\theta-\phi) + i\sin(\theta-\phi))$$

$$+ \frac{2}{3} \times \frac{3}{4}[\cos(\theta-\phi) - i\sin(\theta-\phi)]$$

$$z = \left(2+\frac{1}{2}\right)\cos(\theta-\phi) + i\left(2-\frac{1}{2}\right)\sin(\theta-\phi)$$

$$\therefore \quad |z| = \sqrt{\frac{25}{4}\cos^2(\theta-\phi) + \frac{9}{4}\sin^2(\theta-\phi)}$$

$$= \sqrt{\frac{16}{4}\cos^2(\theta-\phi) + \frac{9}{4}}$$

$$\Rightarrow \quad \frac{3}{2} \le |z| \le \frac{5}{2}$$

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Q4

(1) $z=\left(\frac{\sqrt{3}}{2}+\frac{i}{2}\right)^5+\left(\frac{\sqrt{3}}{2}-\frac{i}{2}\right)^5$

$=\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)^5+\left(\cos\frac{\pi}{6}-i\sin\frac{\pi}{6}\right)^5$

$=\left(e^{i\frac{\pi}{6}}\right)^5+\left(e^{-i\frac{\pi}{6}}\right)^5=2\cos\frac{\pi}{6}=\sqrt{3}$

$\Rightarrow$ $I(z)=0,\ \text{Re}(z)=\sqrt{3}$

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Q5

(1) $-(6+i)^3=x+iy$

$\Rightarrow$ $-[216+i^3+18i(6+i)]=x+iy$

$\Rightarrow$ $-[216-i+108i-18]=x+iy$

$\Rightarrow$ $-216+i-108i+18=x+iy$

$\Rightarrow$ $-198-107i=x+iy$

$\Rightarrow$ $x=-198,\ y=-107$

$\Rightarrow$ $y-x=-107+198=91$

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Q6

(2) Since, $|z|+z=3+i$

Let $z=a+ib$, then

$|z|+z=3+i\Rightarrow\sqrt{a^2+b^2}+a+ib=3+i$

Compare real and imaginary coefficients on both sides

$b=1,\ \sqrt{a^2+b^2}+a=3$

$\sqrt{a^2+1}=3-a$

$a^2+1=a^2+9-6a$

$6a=8$

$a=\frac{4}{3}$

Then,

$|z|=\sqrt{\left(\frac{4}{3}\right)^2+1}=\sqrt{\frac{16}{9}+1}=\frac{5}{3}$

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Q7

(1) Let $t=\frac{z-\alpha}{z+\alpha}$

$\because$ t is purely imaginary number.

$\therefore$ $t+\bar{t}=0$

$\Rightarrow$ $\frac{z-\alpha}{z+\alpha}+\frac{\bar{z}-\alpha}{\bar{z}+\alpha}=0$

$\Rightarrow$ $(z-\alpha)(\bar{z}+\alpha)+(\bar{z}-\alpha)(z+\alpha)=0$

$\Rightarrow$ $z\bar{z}-\alpha^2+z\bar{z}-\alpha^2=0$

$\Rightarrow$ $z\bar{z}-\alpha^2=0$

$\Rightarrow$ $|z|^2-\alpha^2=0$

$\Rightarrow$ $\alpha^2=4$

$\Rightarrow$ $\alpha=\pm 2$

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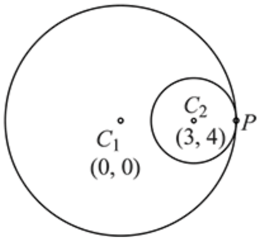
Q8

(1) $|z_1| = 9, |z_2 - 3 - 4i| = 4$

z_1 lies on a circle with centre $C_1(0, 0)$ and radius $r_1 = 9$

z_2 lies on a circle with centre $C_2(3, 4)$ and radius $r_2 = 4$

So, minimum value of $|z_1 - z_2|$ is zero at point of contact (i.e. A)



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Q9

(4) $\frac{\sqrt{3}}{2} + \frac{i}{2} = -i\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = -i\omega$

where ω is imaginary cube root of unity.

Now, $(1 + iz + z^5 + iz^8)^9$

$= (1 + \omega - i\omega^2 + i\omega^2)^9 = (1 + \omega)^9$

$= (-\omega^2)^9 = -\omega^{18} = -1 \qquad (\because\ 1 + \omega + \omega^2 = 0)$

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Q10

(2) Let $z \in S$ then $z = \frac{\alpha + i}{\alpha - i}$

Since, z is a complex number and let $z = x + iy$

Then, $x + iy = \frac{(\alpha + i)^2}{\alpha^2 + 1}$ (by rationalisation)

$\Rightarrow x + iy = \frac{(\alpha^2 - 1)}{\alpha^2 + 1} + \frac{i(2\alpha)}{\alpha^2 + 1}$

Then compare both sides

$x = \frac{\alpha^2 - 1}{\alpha^2 + 1} \qquad ...(1)$

$y = \frac{2\alpha}{\alpha^2 + 1} \qquad ...(2)$

Now squaring and adding equations (1) and (2)

$\Rightarrow x^2 + y^2 = \frac{(\alpha^2 - 1)^2}{(\alpha^2 + 1)^2} + \frac{4\alpha^2}{(\alpha^2 + 1)^2} = 1$

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Q11

(3) $\omega = \frac{5 + 3z}{5 - 5z} \Rightarrow 5\omega - 5\omega z = 5 + 3z$

$\Rightarrow 5\omega - 5 = z(3 + 5\omega) \Rightarrow z = \frac{5(\omega - 1)}{3 + 5\omega}$

$\because |z| < 1, \therefore 5|\omega - 1| < |3 + 5\omega|$

$\Rightarrow 25(\omega\bar{\omega} - \omega - \bar{\omega} + 1) < 9 + 25\omega\bar{\omega} + 15\omega + 15\bar{\omega}$

$(\because |z|^2 = z\,\bar{z})$

$\Rightarrow 16 < 40\omega + 40\bar{\omega} \Rightarrow \omega + \bar{\omega} > \frac{2}{5} \Rightarrow 2\,\text{Re}(\omega) > \frac{2}{5}$

$\Rightarrow \text{Re}(\omega) > \frac{1}{5}$

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Q12

(1) $z = \frac{(1+i)^2}{a - i} \times \frac{a + i}{a + i}$

$z = \frac{(1 - 1 + 2i)(a + i)}{a^2 + 1} = \frac{2ai - 2}{a^2 + 1}$

$|z| = \sqrt{\left(\frac{-2}{a^2 + 1}\right)^2 + \left(\frac{2a}{a^2 + 1}\right)^2} = \sqrt{\frac{4 + 4a^2}{(a^2 + 1)^2}}$

$\Rightarrow |z| = \sqrt{\frac{4(1 + a^2)}{(1 + a^2)^2}} = \frac{2}{\sqrt{1 + a^2}} \qquad ...(1)$

Since, it is given that $|z| = \sqrt{\frac{2}{5}}$

Then, from equation (1),

$\sqrt{\frac{2}{5}} = \frac{2}{\sqrt{1 + a^2}}$

Now, square on both side; we get

$\Rightarrow \frac{2}{5} = \frac{4}{1 + a^2} \Rightarrow 1 + a^2 = 10 \Rightarrow a = \pm 3$

Since, it is given that $a > 0 \Rightarrow a = 3$

Then, $z = \frac{(1+i)^2}{a - i} = \frac{1 + i^2 + 2i}{3 - i} = \frac{2i}{3 - i}$

$= \frac{2i(3 + i)}{10} = \frac{-1 + 3i}{5}$

Hence, $\bar{z} = \frac{-1}{5} - \frac{3}{5}i$

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Q13

(3) Given $|z\omega| = 1$ ...(i)

and $\arg\left(\frac{z}{\omega}\right) = \frac{\pi}{2}$ ...(ii)

$\therefore \frac{z}{\omega} + \frac{\bar{z}}{\bar{\omega}} = 0$ $\left[\because \text{Re}\left(\frac{z}{\omega}\right) = 0\right]$

$\Rightarrow z\bar{\omega} = -\bar{z}\omega$

from equation (i), $z\bar{z}\omega\bar{\omega} = 1$ [using $z\bar{z} = |z|^2$]

$(\bar{z}\omega)^2 = -1 \Rightarrow \bar{z}\omega = \pm i$

from equation (ii), $-\arg(\bar{z}) - \arg\omega = \frac{\pi}{2}$ $-\arg(\bar{z}\omega) = \frac{-\pi}{2}$

Hence, $\bar{z}w = -i$

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Q14

(2) Given equation is, $|z - 1| = |z - i|$

$\Rightarrow$ $(x - 1)^2 + y^2 = x^2 + (y - 1)^2$ [Here, $z = x + iy$]

$\Rightarrow$ $1 - 2x = 1 - 2y \Rightarrow x - y = 0$

Hence, locus is straight line with slope 1.

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Q15

(3) Let Re $(z) = x$ i.e., $z = x + 10i$

$2z - n = (2i - 1)(2z + n)$

$(2x - n) + 20i = (2i - 1)((2x + n) + 20i)$

On comparing real and imaginary parts,

$-(2x + n) - 40 = 2x - n$ and $20 = 4x + 2n - 20$

$\Rightarrow 4x = -40$ and $40 = -40 + 2n$

$\Rightarrow x = -10$ and $n = 40$

Hence, Re$(z) = -10$

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